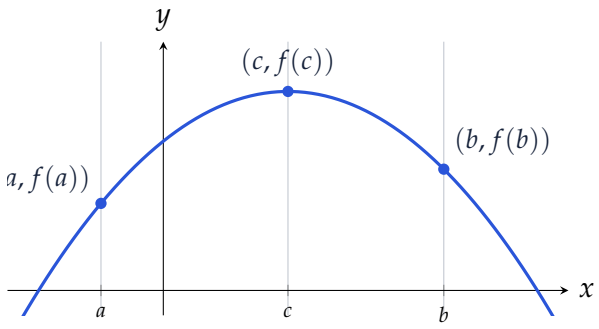
Lesson 1-5 • IVT and EVT

Notes (filled in)

Objective: use the Intermediate Value Theorem and the Extreme Value Theorem, and know what each one needs.

Symbolic notation of graphs



A point on the graph is written $(x, f(x))$: $f(a)$ is the height of the graph at $x = a$.

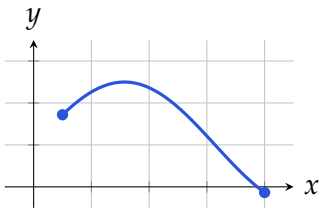
Maximum at $x = c$:
$f(c) \geq f(x)$ for every x in the interval

one height, at $x = c$, is above all the other heights

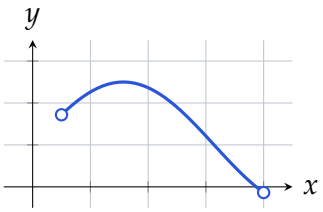
Minimum at $x = c$:
$f(c) \leq f(x)$ for every x in the interval

one height is below all the others

Types of intervals



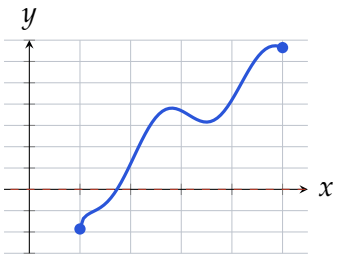
Closed interval: endpoints included
inequality $a \leq x \leq b$, interval $[a, b]$



Open interval: endpoints excluded
inequality $a < x < b$, interval (a, b)

Intermediate Value Theorem (IVT)

Needs	If f is __continuous__ on $[a,b]$,
Proves	then f takes every __height (value)__ between $f(a)$ and $f(b)$ at least __once__.



f is continuous on $[1,5]$, $f(1) < 0 < f(5)$.
The IVT guarantees a value c in __(1,5)__ with $f(c) =$ __0__: the graph cannot go from below the x-axis to above it without __crossing__ it.

Try it. g is continuous.

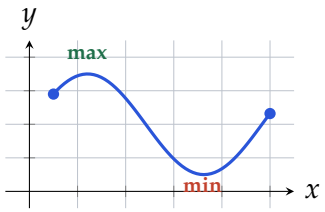
x	0	2	4	6
$g(x)$	3	-1	5	8

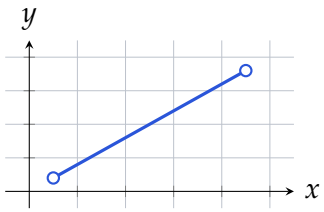
At least how many zeros does g have on $[0,6]$?

At least 2: g changes sign on $[0,2]$ (3 to -1) and on $[2,4]$ (-1 to 5). The IVT gives a zero in each interval.

Extreme Value Theorem (EVT)

Needs	If f is __continuous__ on a __closed__ interval $[a,b]$,
Proves	then f has an absolute __maximum__ and an absolute __minimum__ on $[a,b]$.





continuous on $[a,b]$: max and min __exist__ open interval: __no__ max and __no__ min
A horizontal segment on $[a,b]$ is a special case: every point is both a maximum and a minimum.

Summary

IVT needs continuity on $[a,b]$ and proves that every height between $f(a)$ and $f(b)$ is reached. EVT needs continuity on a closed interval and proves that an absolute maximum and minimum exist.